TASK MANAGER APPLICATION

Test Execution Report

Backend API • Authentication • Task Validation • Security Checks

1. Report Details

Project	Task Manager Application
Role	Test Engineer
Prepared by	Shaikh Abdullah Mahdi
Test area	Backend REST API
Execution environment	Local development environment; Node.js, Jest, Supertest, PostgreSQL, Postman
Git branch	feature/abdullah-api-security-tests
Report status	Draft based on the test output and Postman results shared

2. Objective

The objective of this execution was to check the backend API's authentication behaviour, task input validation, session handling, and protection of user-specific task data. The report records the automated Jest results and the manual/API checks performed through Postman.

3. Test Environment

- Backend application: Node.js and Express.
- Automated test runner: Jest, executed with `npm.cmd test` from the backend directory.
- HTTP test client for automated tests: Supertest.
- Database: PostgreSQL using a dedicated local test database (`taskmanager_test`).
- Manual API testing: Postman, using the local backend base URL `http://localhost:3000`.
- Health endpoint used for a basic availability check: `GET /health`.

4. Automated Test Execution Summary

Test suites	1 passed / 1 total
Tests	21 passed / 21 total
Snapshots	0
Reported execution time	28.948 seconds
Overall result	PASS

Evidence: the provided Jest console output showed `PASS tests/auth.test.js`, `Test Suites: 1 passed, 1 total`, and `Tests: 21 passed, 21 total`.

5. Automated Test Coverage

Registration	Valid registration creates a user; invalid email and weak password are rejected; duplicate registration is rejected even when email case differs; response does not expose the password hash.
Login	Valid login returns a session token; incorrect credentials are rejected; missing email/password is rejected; response does not expose the password hash.

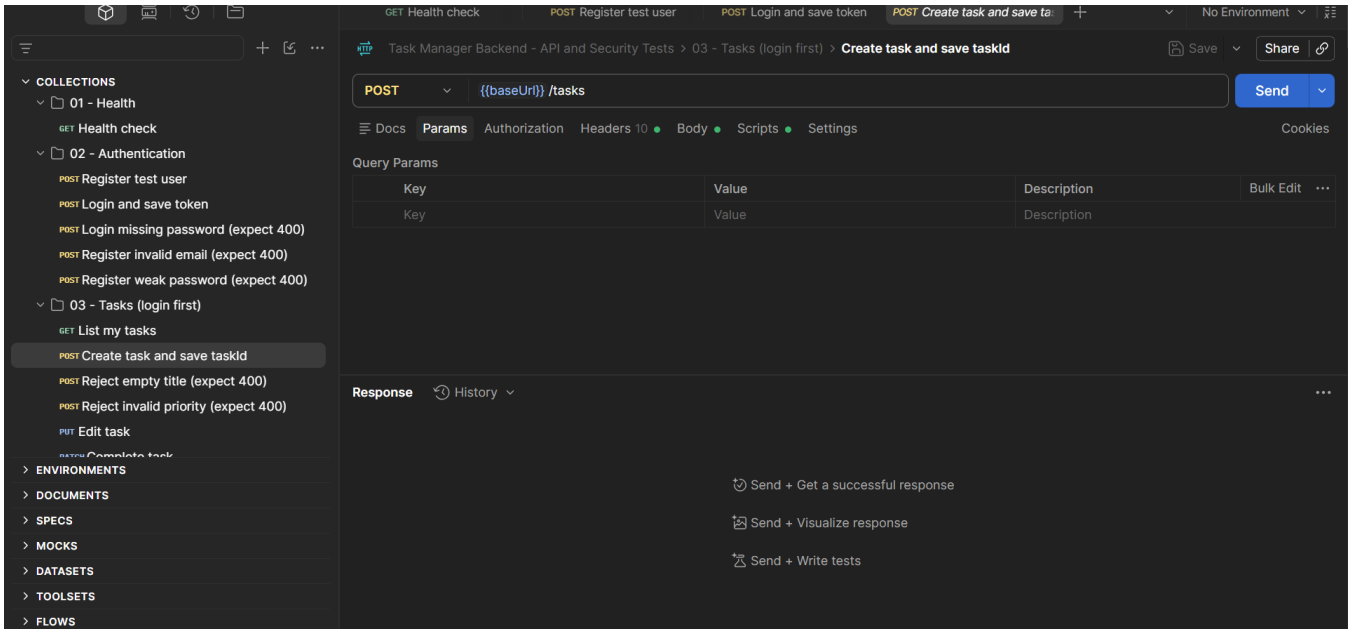
Logout and sessions	Logout invalidates the session; reuse of an invalidated session is rejected; an idle session beyond 30 minutes is rejected.
Password change	Incorrect current password is rejected; successful password change invalidates existing sessions.
Authentication enforcement	Protected task operations reject requests without an authentication token.
Task ownership and isolation	A user cannot edit or delete another user's task; users only see their own tasks.
Task input validation	Empty task title, invalid priority, and invalid due date are rejected.

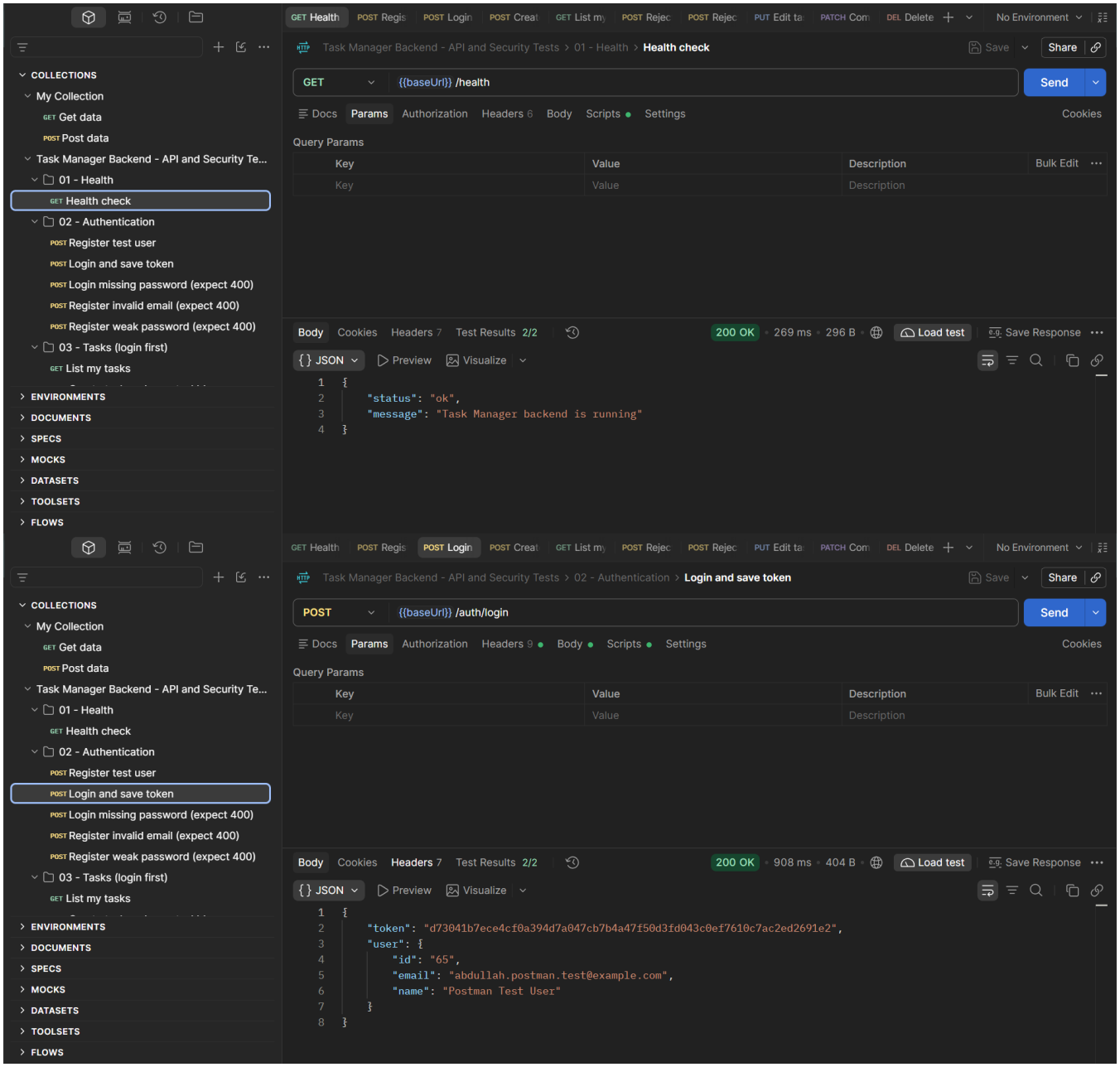
6. Postman API Checks

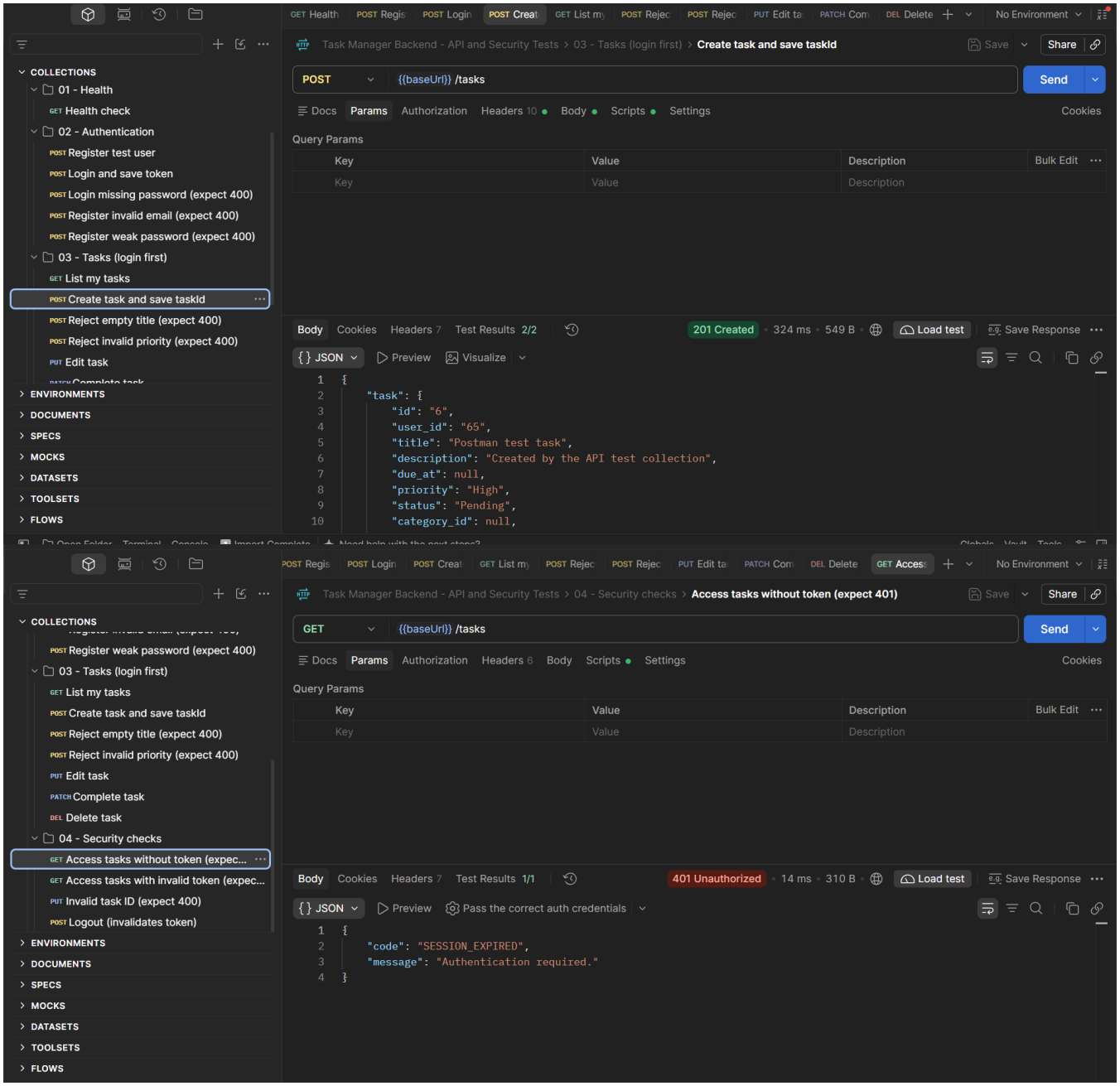
Based on the results reported during testing, the following checks were performed with the local backend:

- `GET /health` returned HTTP 200.
- `POST /auth/register` and `POST /auth/login` were tested; login returned HTTP 200.
- Task creation returned HTTP 201.
- Requests in the Postman sections `03 - Tasks (login first)` and `04 - Security checks` were run, and the user reported that the returned status codes matched the collection's expected results.
- A request to `GET /tasks` without a token returned HTTP 401, confirming that the mounted task route required authentication.

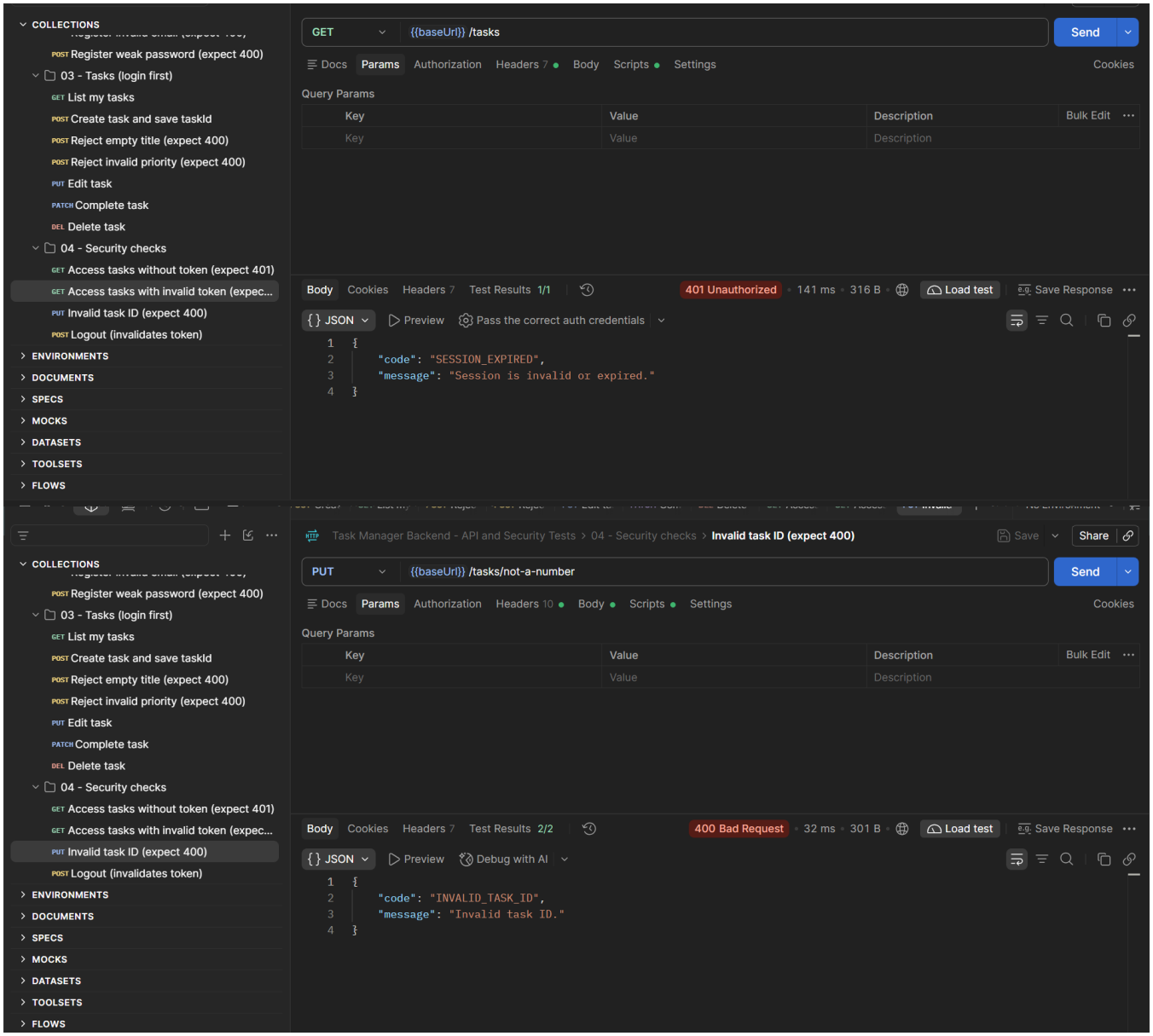
Note: individual response screenshots and a per-request Postman export were not included with the information used to prepare this report. Attach screenshots or export the Postman run results if the faculty requires direct evidence.

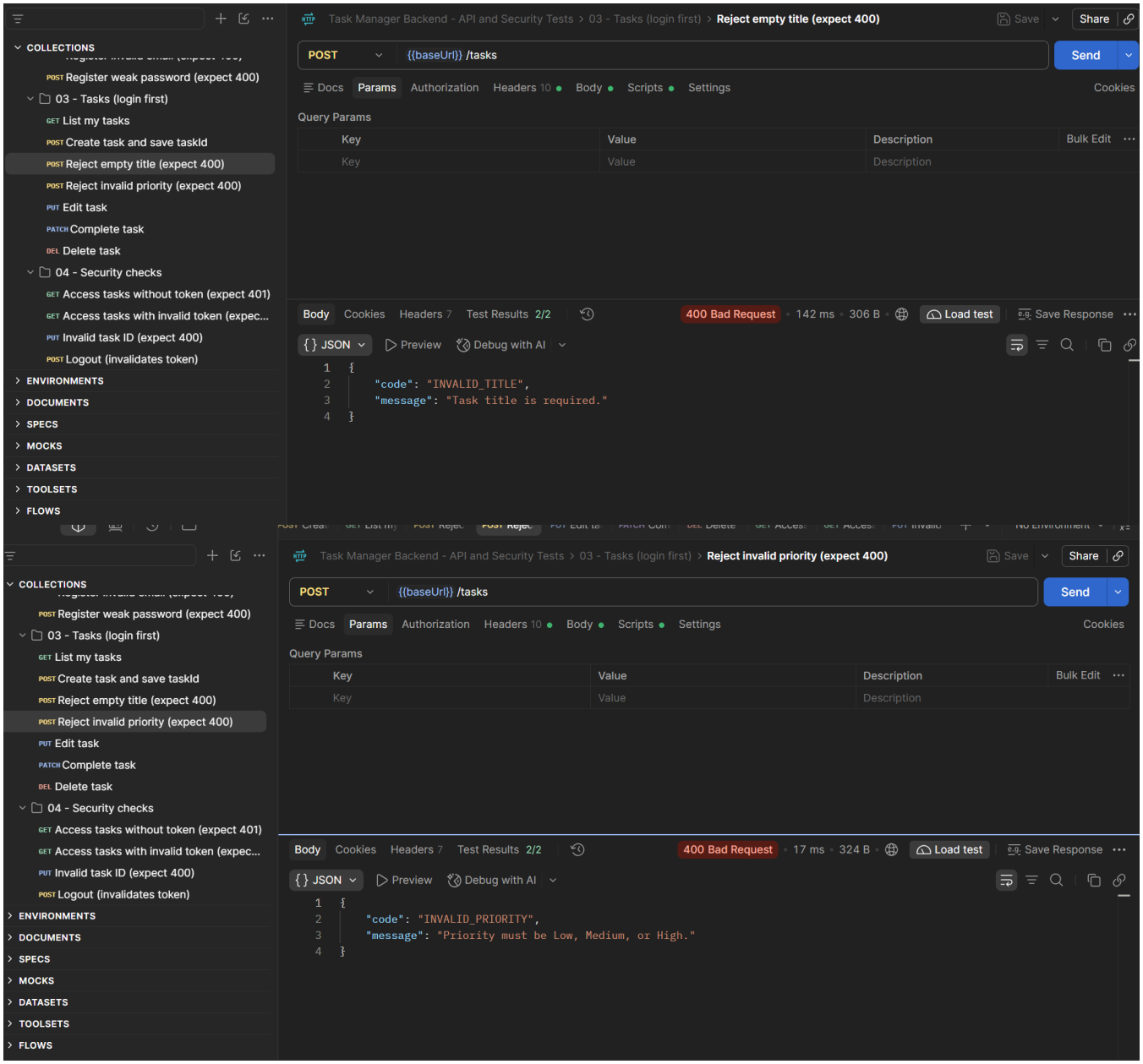



COLLECTIONS
My Collection
GET Get data
POST Post data
Task Manager Backend - API and Security Te...
01 - Health
GET Health check
02 - Authentication
POST Register test user
POST Login and save token
POST Login missing password (expect 400)
POST Register invalid email (expect 400)
POST Register weak password (expect 400)
03 - Tasks (login first)
GET List my tasks
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
No Environment
Task Manager Backend - API and Security Tests › 01 - Health › Health check
Save
Share
GET
{{baseUrl}} /health
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 7
Test Results 2/2
200 OK
269 ms
296 B
Load test
Save Response
{ } JSON
Preview
Visualize
1 {
2 "status": "ok",
3 "message": "Task Manager backend is running"
4 }
Task Manager Backend - API and Security Tests › 02 - Authentication › Login and save token
POST
{{baseUrl}} /auth/login
Headers 9
Body
Scripts
Settings
908 ms
404 B
1 {
2 "token": "d73041b7ece4cf0a394d7a047cb7b4a47f50d3fd043c0ef7610c7ac2ed2691e2",
3 "user": {
4 "id": "65",
5 "email": "abdullah.postman_test@example.com",
6 "name": "Postman Test User"
7 }
8 }


COLLECTIONS
01 - Health
Health check
02 - Authentication
Register test user
Login and save token
Login missing password (expect 400)
Register invalid email (expect 400)
Register weak password (expect 400)
03 - Tasks (login first)
List my tasks
Create task and save taskId
Reject empty title (expect 400)
Reject invalid priority (expect 400)
Edit task
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
No Environment
Task Manager Backend - API and Security Tests › 03 - Tasks (login first) › Create task and save taskId
Save
Share
POST
{{baseUrl}} /tasks
Send
Docs
Params
Authorization
Headers 10
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 7
Test Results 2/2
201 Created
324 ms
549 B
Load test
Save Response
JSON
Preview
Visualize
{
"task": {
"id": "6",
"user_id": "65",
"title": "Postman test task",
"description": "Created by the API test collection",
"due_at": null,
"priority": "High",
"status": "Pending",
"category_id": null,

COLLECTIONS
Register weak password (expect 400)
03 - Tasks (login first)
List my tasks
Create task and save taskId
Reject empty title (expect 400)
Reject invalid priority (expect 400)
Edit task
Complete task
Delete task
04 - Security checks
Access tasks without token (expec...
Access tasks with invalid token (expec...
Invalid task ID (expect 400)
Logout (invalidates token)
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
No Environment
Task Manager Backend - API and Security Tests › 04 - Security checks › Access tasks without token (expect 401)
Save
Share
GET
{{baseUrl}} /tasks
Send
Docs
Params
Authorization
Headers 6
Body
Scripts
Settings
Cookies
Query Params
Key
Value
Description
Bulk Edit
Body
Cookies
Headers 7
Test Results 1/1
401 Unauthorized
14 ms
310 B
Load test
Save Response
JSON
Preview
Pass the correct auth credentials
{
"code": "SESSION_EXPIRED",
"message": "Authentication required."
}


COLLECTIONS
POST Register weak password (expect 400)
03 - Tasks (login first)
GET List my tasks
POST Create task and save taskId
POST Reject empty title (expect 400)
POST Reject invalid priority (expect 400)
PUT Edit task
PATCH Complete task
DEL Delete task
04 - Security checks
GET Access tasks without token (expect 401)
GET Access tasks with invalid token (expec...
PUT Invalid task ID (expect 400)
POST Logout (invalidates token)
ENVIRONMENTS
DOCUMENTS
SPECS
MOCKS
DATASETS
TOOLSETS
FLOWS
GET {{baseUrl}} /tasks
Send
Docs Params Authorization Headers 7 Body Scripts Settings Cookies
Query Params
Key Value Description Bulk Edit
Body Cookies Headers 7 Test Results 1/1
401 Unauthorized 141 ms 316 B
Load test Save Response
JSON Preview Pass the correct auth credentials
{
"code": "SESSION_EXPIRED",
"message": "Session is invalid or expired."
}
Task Manager Backend - API and Security Tests > 04 - Security checks > Invalid task ID (expect 400)
Save Share
PUT {{baseUrl}} /tasks/not-a-number
Send
Docs Params Authorization Headers 10 Body Scripts Settings Cookies
Query Params
Key Value Description Bulk Edit
Body Cookies Headers 7 Test Results 2/2
400 Bad Request 32 ms 301 B
Load test Save Response
JSON Preview Debug with AI
{
"code": "INVALID_TASK_ID",
"message": "Invalid task ID."
}



7. Route Mounting Fix

During local verification, `backend/server.js` was updated to mount the task router with `app.use("/tasks", require("./routes/tasks"));`. After the change, the health endpoint returned HTTP 200 and an unauthenticated request to `GET /tasks` returned HTTP 401. The fix was committed on the feature branch.

8. Issues and Observations

- The Jest output included dotenv messages about loading environment variables. These were logs, not test failures; the suite completed successfully.
- The report records the behaviour covered by the 21 passing tests, not proof of full application readiness.
- Reminder scheduling, push notifications, backup/restore scripts, and their integration were not demonstrated by the supplied test output; their implementation and status must be verified separately.

9. Conclusion

The backend automated test suite passed all 21 tests. The tested areas include authentication, session invalidation, task validation, task ownership, and user-data isolation. Basic Postman checks also produced the reported expected HTTP status codes. Further evidence should be attached for formal submission, and unverified deliverables should remain marked as pending until tested.

10. Suggested Evidence to Attach

- Screenshot of the terminal showing `21 passed, 21 total`.
- Postman screenshot or exported run results for health, registration/login, task creation, task operations, and security checks.
- Screenshot of `git log -4 --oneline` showing the testing commits on `feature/abdullah-api-security-tests`.

